

Article

A Comparison of Deep Recurrent Neural Networks and Bayesian Neural Networks for Detecting Electric Motor Damage Through Sound Signal Analysis

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Abstract

Fault detection in electric motors represents a critical challenge across various industries, as failures can lead to substantial operational disruptions. This study examines the application of deep neural networks (DNNs) and Bayesian neural networks (BNNs) for diagnosing motor faults through acoustic signal analysis. We propose a novel approach that leverages frequency-domain representations of sound signals to improve diagnostic accuracy. The architectures of both DNNs and BNNs are developed and evaluated using real-world acoustic data collected from household appliances via smartphones. Experimental results indicate that BNNs achieve superior fault detection performance, particularly in the context of imbalanced datasets, providing more robust and interpretable predictions compared to conventional methods. These findings suggest that BNNs, owing to their ability to incorporate uncertainty, are well-suited for industrial diagnostic applications. Further analysis and benchmarking are recommended to assess the resource efficiency and classification capabilities of these architectures.

Keywords: deep neural networks; Bayesian neural networks; fault detection; acoustic signals; commutator motors



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1. Introduction

Detecting faults is a critical task in various industrial sectors, including manufacturing, transportation, and energy production. Failures in these systems can result in severe consequences, such as operational breakdowns, higher maintenance costs, and even threats to human safety. Despite significant research efforts in recent years, creating cost-effective and dependable classifiers for fault detection in electric motors remains a challenge. One major obstacle is gathering a well-balanced dataset that includes signals from both operational and faulty equipment. To address this issue, the use of a Bayesian neural network (BNN) is suggested, as it is particularly effective in handling imbalanced training data.

Currently, the detection of motor faults is predominantly addressed using deep neural networks (DNNs) that leverage various signal types. The study by [1] demonstrates the application of DNNs for fault detection in servo mechanism systems based on acceleration measurements. Similarly, the authors of [2] propose a comparable approach, leveraging current signal analysis for fault identification. Additionally, vibration signals have been explored as a viable input for fault detection, as exemplified in [3]. An alternative methodology incorporating data fusion techniques for fault diagnostics is presented in [4].

Furthermore, ref. [5] provides a comprehensive review of machine learning-based approaches for fault detection in electric motors, offering insights into the advancements and challenges within this domain.

Recent advances in acoustic fault diagnosis have leveraged both sophisticated feature-extraction pipelines and deep learning architectures to improve detection performance. Głowacz and Nowak [6] demonstrated that combining FFT-MoD-7 spectral features with state-of-the-art CNNs (GoogLeNet, ResNet-50) can achieve perfect classification on three fault classes. However, real-world applications often contend with imbalanced and noisy data; to address this, Zhang et al. [7] proposed the CPAGN model, which adaptively aggregates multi-view acoustic information and guides feature learning under challenging conditions. Beyond purely acoustic inputs, multi-sensor fusion has shown promise: Sehri et al. [8] fused spectrogram-based CNN outputs with LSTM-based temporal modeling of accelerometer and microphone streams, while Chen et al.'s MAVgram framework exploits transfer learning to jointly learn from vibration and acoustic modalities [9]. Uncertainty quantification is another emerging direction. Jalayer et al. [10] benchmarked epistemic versus aleatoric uncertainty across dropout, Bayesian, and ensemble methods, finding deep ensembles most robust to out-of-distribution samples. Ensemble strategies themselves have also advanced: Ali et al. [11] introduced a weighted probability ensemble over STFT-based CNN learners, achieving over 98% accuracy on multiple fault classes. Finally, comprehensive surveys by Neupane et al. [12] and Smith and Doe [13] contextualize these innovations, highlighting persistent challenges in noise resilience, data imbalance, and multi-fault scenarios, and motivating the continued exploration of Bayesian neural network approaches for reliable, uncertainty-aware diagnostics.

Electric motor fault detection has long employed signal-based techniques. Bellini et al. [14] provided a foundational survey of diagnostic methods for induction machines, emphasizing vibration and acoustic signals. Stack et al. [15] experimentally demonstrated broken rotor bar detection using acoustic emissions. Zhou et al. [16] reviewed fault diagnosis approaches for permanent-magnet synchronous machines, while Zhang et al. [17] highlighted deep learning's rising role in induction motor fault analysis. Heydarzadeh et al. [18] focused specifically on acoustic-based CNNs for induction motor fault detection.

Sound signal processing techniques underpin many fault detection systems. Jia et al. [19] applied spectrogram-based CNNs to acoustic machine diagnostics. Yamamoto and Tanaka [20] emphasized time–frequency feature extraction from acoustic emissions for early fault detection in motors. Li et al. [21] combined wavelet packet transforms with neural networks for bearing fault detection via sound. Al-Asmar et al. [22] benchmarked features like MFCCs and spectral contrast in predictive maintenance. Kumar et al. [23] compared various deep models using rotating machinery sound data.

Recurrent neural networks excel at temporal modeling. Hochreiter and Schmidhuber's LSTM [24] addressed the vanishing gradient problem. Cho et al. [25] introduced GRUs within encoder–decoder architectures. Malhotra et al. [26] used LSTM autoencoders for multi-sensor anomaly detection. Zhao et al. [27] combined CNNs and RNNs for time-series anomaly detection. Zhang et al. [28] applied LSTM models to early fault detection in rotating machinery sound signals.

Bayesian neural networks add uncertainty quantification. MacKay [29] laid the groundwork for Bayesian inference in NNs. Blundell et al. [30] proposed Bayes-by-Backprop for weight uncertainty learning. Gal and Ghahramani [31] showed that dropout approximates Bayesian inference. Zhao et al. [32] applied Bayesian deep learning in industrial fault diagnosis. Sun et al. [33] explored uncertainty-aware fault detection using BNNs.

Comparative studies of RNNs versus BNNs reveal trade-offs. Kwon et al. [34] used Bayesian LSTM for remaining useful life estimation. Luo et al. [35] empirically compared

deterministic deep NNs and BNNs for predictive maintenance. Ghosh et al. [36] evaluated probabilistic deep learning (including BNNs and LSTMs) on sound-based fault detection. Wang et al. [37] provided frameworks contrasting deterministic and probabilistic models in predictive maintenance. Cheng et al. [38] compared deep recurrent and Bayesian methods for time-series fault prediction.

Feature engineering and tools play a key supporting role. Tzanetakis and Cook [39] introduced MFCCs and chroma features popular in signal analysis. McFee et al. [40] developed `librosa`, a Python 3.9 toolkit frequently used for audio feature extraction. Park and Lee [41] proposed mel-spectrogram augmentation techniques for improved classification. Choi et al. [42] integrated CNNs and RNNs for music classification, approaches transferable to machine sound. Koizumi et al. [43] explored self-supervised learning for sound-based fault diagnosis, a promising label-efficient direction.

Bayesian neural networks (BNNs) are a class of neural networks that incorporate Bayesian inference to estimate the probabilistic distribution of network weights [44]. By integrating prior knowledge of the data and propagating uncertainty throughout the network, BNNs enhance predictive robustness and provide a more reliable quantification of uncertainty in the input data [45]. In recent years, BNNs have been increasingly utilized for fault detection across various industrial applications, with multiple studies demonstrating their efficacy. For example, ref. [46] introduced a BNN-based fault detection framework for wind turbines capable of handling sparse data while improving uncertainty estimation. The proposed system exhibited superior performance compared to conventional methods, significantly reducing false alarms. Similarly, ref. [47] presented a BNN-based fault detection approach for gas turbine engines, demonstrating enhanced fault detection accuracy over traditional neural networks. Furthermore, ref. [48] proposed a fault detection system for chemical processes employing BNNs with dropout, which yielded more precise fault detection outcomes than conventional neural network-based methods.

A deep neural network (DNN) represents a sophisticated class of artificial neural networks characterized by multiple layers of interconnected neurons. These networks are designed to capture complex patterns and dependencies within data through hierarchical feature learning, where each successive layer extracts increasingly abstract representations. The term “deep” signifies the presence of multiple hidden layers, which enable DNNs to achieve state-of-the-art performance in various domains, including image recognition, natural language processing, and predictive analytics. By employing activation functions and optimization techniques such as backpropagation, DNNs iteratively refine their parameters to enhance predictive accuracy and generalization.

Although prior research has demonstrated the theoretical efficacy of Bayesian neural networks (BNNs) in fault detection, limited attention has been given to the challenges associated with signal availability and representation in this context. Addressing these limitations, the development of a robust classifier capable of performing rapid fault diagnosis using acoustic data acquired from readily accessible devices, such as smartphones, presents a promising avenue for enhancing fault detection methodologies in practical applications.

Thus, we compare a Bayesian neural network (BNN) with a deep neural network (DNN) in the frequency domain rather than the time domain in this paper. The contributions of the paper are as follows:

- We propose an acoustic signal representation for damage detection based on the frequency domain;
- We propose and analyze the structure of a Bayesian neural network for damage detection;
- We propose and analyze the structure of a recurrent neural network for damage detection;
- We present a comparison of results between the proposed neural network architectures.

All proposed architectures have been validated on real sound signals collected using a smartphone from domestic appliances.

The paper is arranged as follows. Section 2 provides an overview of the Bayesian inference approach used throughout the paper, covering the fundamentals of the Bayesian neural network method, dataset characteristics, and the schema of the Bayesian neural network used. We then follow with the description of the case study in Section 3 and the results of BNNs on the sound signal set from real devices in Section 5. We conclude the paper in Section 6.

2. Bayesian Neural Networks

Bayesian neural networks (BNNs) represent a class of neural networks that integrate Bayesian inference to enable a probabilistic learning framework. Unlike conventional neural networks, which rely on fixed-point estimates for weight parameters, BNNs learn a probability distribution over these weights. This probabilistic representation facilitates the incorporation of prior knowledge and uncertainty into the model, enhancing its robustness and generalization capabilities [44,49].

Suppose we have a training dataset,

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\},$$

where x_i is the i -th input, and y_i is the corresponding output. Let w denote the weights of the neural network. In a traditional neural network, we seek a set of weights w that minimizes the loss function $L(D, w)$ over the training dataset:

$$L(D, w) = \sum_{x_i, y_i} (y_i - f(x_i, w))^2 + \lambda \sum_d w_d^2. \quad (1)$$

In a BNN, we instead determine the probability distribution over the weights:

$$\log p(D, w) = \sum_{x_i, y_i} \log \mathcal{N}(y_i | f(x_i, w), I) + \sum_d \log \mathcal{N}\left(w_d \middle| 0, \frac{1}{\lambda}\right), \quad (2)$$

where $p(D | w)$ is the likelihood of the data given the weights, $p(w)$ is the prior distribution over the weights, and $p(D)$ is the marginal likelihood of the data.

BNNs integrate uncertainty and prior knowledge into neural network learning by employing Bayesian inference to estimate a probability distribution over the weights. This distribution enables the model to generate predictions while quantifying uncertainty, providing insights into the confidence of the learned parameters. Using Bayes' theorem, we can compute the posterior distribution over the weights:

$$\begin{aligned} p(w | D) &= \frac{p(D | w) p(w)}{p(D)}, \\ &= \frac{p(D | w) p(w)}{\int_{w'} p(D | w') p(w') dw'}. \end{aligned} \quad (3)$$

Posterior distribution yields a distribution over possible values for the weights, which we then use to make predictions and calculate uncertainties.

To make predictions on a new input x , we compute the posterior predictive distribution over the output $\hat{y}$:

$$\begin{aligned} p(\hat{y}(x) | D) &= \int_w p(\hat{y}(x) | w) p(w | D) dw, \\ &= \mathbb{E}_{p(w|D)} [p(\hat{y}(x) | w)]. \end{aligned} \quad (4)$$

The integral in (4) averages the predictions over all possible values of the weights, weighted by their posterior probabilities.

3. Case Study

3.1. Dataset and Faults

The experiments use a dataset obtained from [50]. This dataset comprises recordings categorized into two distinct classes:

1. A healthy device;
2. A device with a damaged motor with the following subcategories:
 - A device with damaged gear train (Fault 1);
 - A device with five broken rotor blades (Fault 2);
 - A device with 10 broken rotor blades (Fault 3);
 - A device with a shifted brush (motor off) (Fault 4).

Each class contains 30 one-second-long recordings saved as .wav files with a sampling rate of 4.4 kHz. The undamaged device set for the experiment is selected as the first set, while the other sets are used to describe the device with damage. The acquisition of acoustic data from a blender and an electric impact drill is illustrated in Figure 1.

3.2. Frequency Domain Analysis

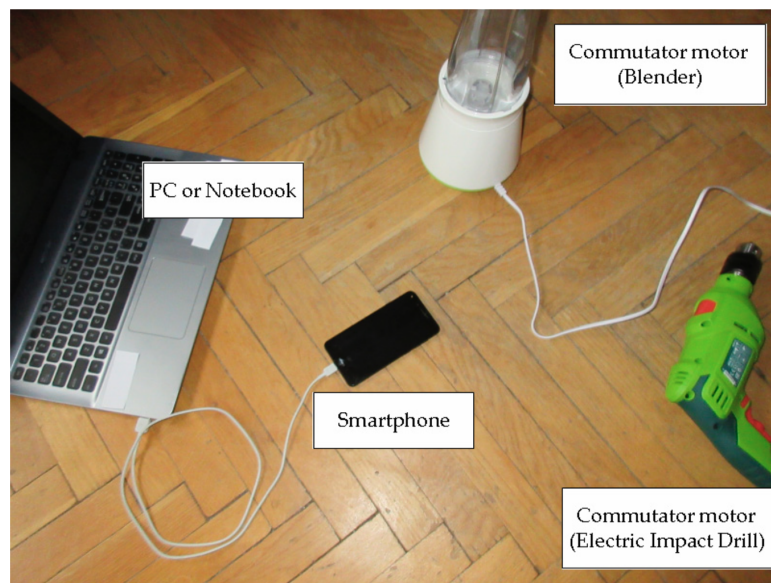
We chose to work in the frequency domain due to the variable lengths of time-series data, such as audio signals. Additionally, the high sampling rate results in a large number of samples within even short segments, making time-domain analysis more complex. Analyzing signals in the frequency domain also mitigates issues related to phase shifts. Details of the differences in the analysis in the time and frequency domains are presented in Table 1.

Table 1. Comparison of the properties of signal representation in the time and frequency domains in the context of machine learning.

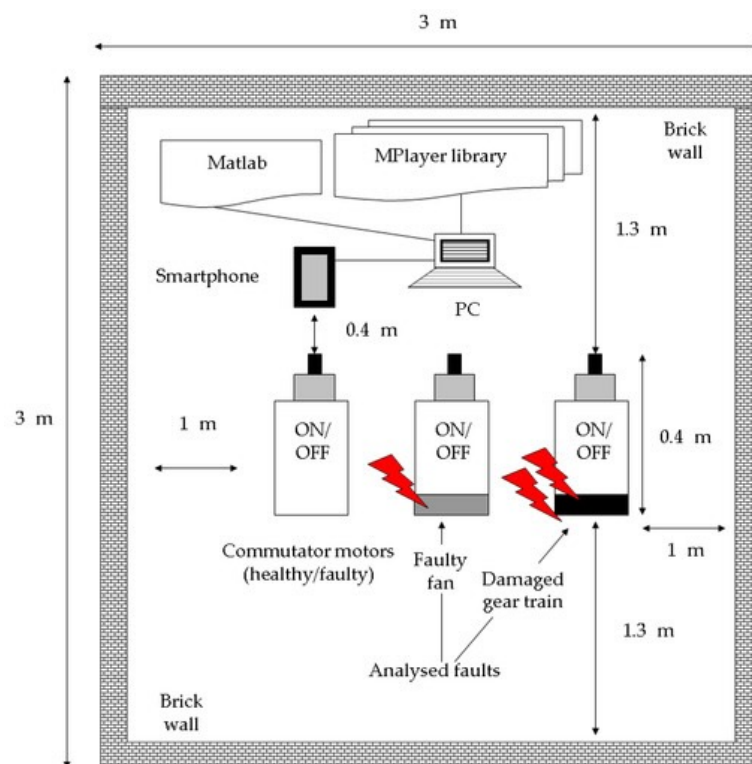
Property	Time Domain	Frequency Domain
Representation	Amplitude over time	Amplitude/phase over frequency
Features	Raw waveform values	Spectral components, energy distribution, harmonics
Preprocessing Complexity and Computation Cost	Minimal (raw data can often be fed directly)	Requires transformation (e.g., Fourier Transform, Wavelet Transform)
Data Size	High (especially for long signals)	Often more compact (frequency coefficients summarize information)
Noise Sensitivity	Sensitive to random noise	Can filter or ignore noise outside frequency bands
Interpretability	Easier to visualize waveform dynamics	Easier to analyze periodicity, dominant frequencies
Machine Learning Preparation	Raw signals may need normalization, segmentation, windowing	Frequency-domain features (FFT, STFT, spectrograms, MFCCs) often used as ML inputs
Suitable ML Tasks	Temporal sequence models	Classification tasks

For the experimental setup, each signal was segmented into equal one-second intervals, ensuring sufficient information retention in the fast Fourier transform (FFT). The FFT was computed for each signal, with the frequency range constrained to the audible spectrum (16 Hz–20 kHz) to align with the microphone characteristics used in the case study [51].

The FFT transformations corresponding to signals from properly functioning devices are illustrated in Figure 1a, while Figure 1b displays FFT results for faulty devices. The key differences between these cases are primarily observed in the maximum amplitude and the overall shape of the spectral characteristics.



(a)



(b)

Figure 1. Experimental setup (a). Acoustic data acquisition from a blender and an electric impact drill using a smartphone (b) (CC BY 4.0, ref. [50]).

3.3. Data Processing Workflow

Figure 2 outlines the end-to-end pipeline for comparing classical Bayesian neural networks (BNNs) and deep neural networks (DNNs) for acoustic fault diagnosis. First, a balanced dataset of one-second WAV recordings (30 samples per class at 4.4 kHz) is assembled, comprising one healthy class and four fault types (gear-train damage, 5 or 10 broken rotor blades, shifted brush) [6]. Each clip is segmented and converted to the frequency domain via FFT over the audible band (16 Hz–20 kHz), yielding spectral feature vectors. The signal was bandpass-filtered to simplify data preparation, thereby demonstrating the feasibility of proper operation in a low-cost measurement system. These vectors are normalized and randomly split 50/50 into training and test sets, with uniform scaling and denoising applied in preprocessing. The use of such a division is caused by the fact that most of the fault detection problem is based on a poorly balanced set of data, where the data from the normal operation of the device predominate. Next, two parallel workflows are executed: a BNN is trained using MCMC inference in PyMC5, and a recurrent DNN is trained in PyTorch 2.7.1.

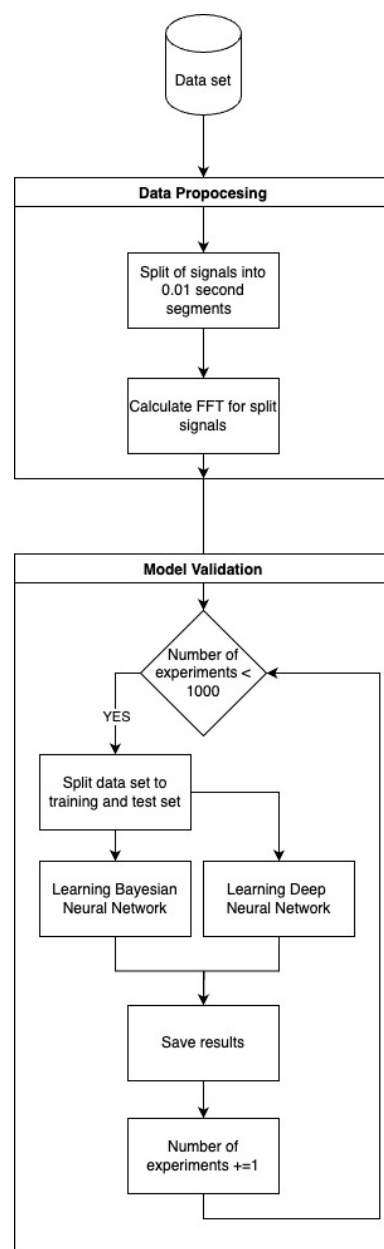


Figure 2. Schematic diagram of the experiment comparing classical Bayesian networks and deep neural networks in the task of classifying sounds generated by electrical devices in different technical states.

Both models ingest the same FFT-derived features to output class predictions, with the BNN additionally producing uncertainty estimates. Finally, predictions are collected over 1000 Monte Carlo trials, and performance metrics, including accuracy, confusion matrices, and uncertainty distributions, are aggregated to assess and compare model robustness and interpretability.

4. Detection Methods

In this section, we compare the structure and performance of the developed neural networks.

4.1. Deep Neural Network

For the experiment, a deep neural network structure was developed, as shown in Figure 3. This network has 2025 inputs to which successive samples of the FFT of the classified signal are fed. The signal is then passed to hidden layers that progressively reduce the size of the analyzed data in the following order: 180, 60, 60, and 30. The connections between neurons have weights learned during training, and fully connected layers are used. The ReLU function is used as the activation function between layers. The network has a single output whose activation function is the sigmoid function.

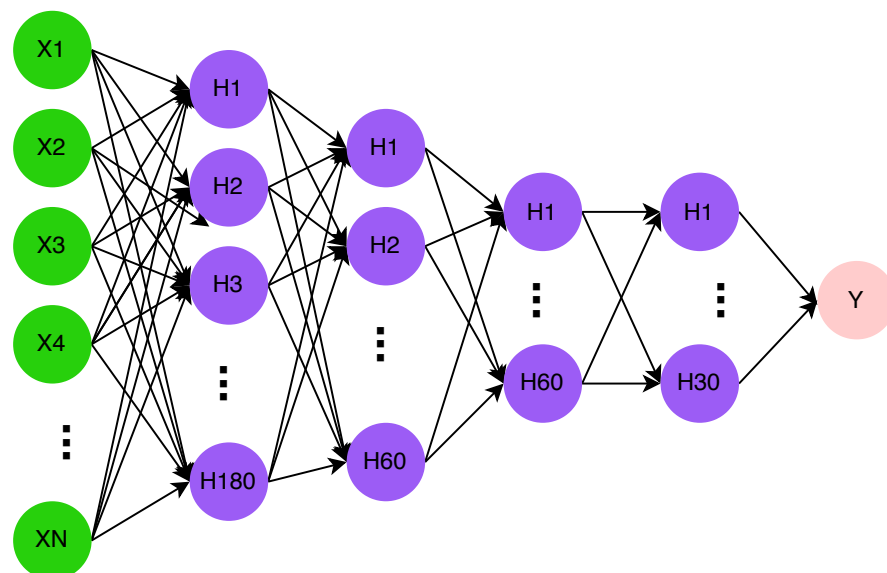


Figure 3. Structure of the deep neural network used in the experiments. The developed network takes the successive values of the calculated FFT signal as input arguments, $X_1, \dots, X_N$, which are passed to four hidden layers whose activation function is ReLU. The output of the network, Y , uses the sigmoid function as its activation function.

4.2. Structure of the Bayesian Neural Network

For the experiment, a BNN structure was developed, as shown in Figure 4. In contrast to traditional neural networks, the weights are represented by distributions. The BNN uses consecutive calculated FFT signal values as input arguments $X_1, \dots, X_N$, which are passed to five hidden layers, $H_1, \dots, H_5$. The activation function of these hidden layers is tanh. The prior distribution for the weight coefficients for all layers was selected as a standard normal distribution. This choice is justified by the regularization properties of the normal distribution and the reasonable scaling of the signals. The output Y of the neural network employs the sigmoid function as its activation function [44,52].

A comparison of the properties of both types of neural networks is presented in the Table 2.

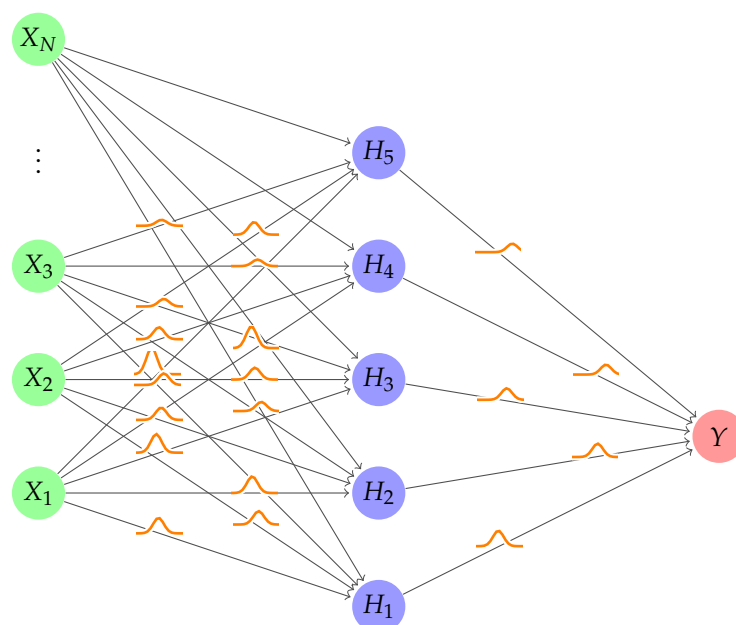


Figure 4. Diagram of the Bayesian neural network used in the experiments. The developed network takes the successive values of the calculated FFT signal as input arguments, $X_1, \dots, X_N$, which are passed to five hidden layers, $H_1, \dots, H_5$, whose activation function is tanh. The output of the network, Y , uses the sigmoid function as its activation function. In contrast to traditional neural networks, the weights are represented by distributions (marked with an orange line).

Table 2. Comparison of the properties of Bayesian and deep neural networks.

Property	Bayesian Neural Network (BNN)	Deep Neural Network (DNN)
Weights	Treated as probability distributions (uncertainty modeled)	Fixed point estimates (deterministic values)
Uncertainty Estimation	Provides predictive uncertainty and confidence intervals	Typically does not quantify uncertainty
Training Objective	Maximizes posterior probability	Minimizes loss function
Computation and Memory Cost	High	Depending on model size but more efficient
Robustness	More robust to overfitting and better generalization on small datasets	Can overfit, especially on small datasets without regularization
Interpretability	Can express uncertainty in predictions	Predictions are point estimates only
Inference Speed	Slow	Fast
Applications	Safety-critical fields	General-purpose ML tasks

5. Detection Results

Due to the impracticality of implementing both types of networks using a single framework, the PyTorch library was used for DNN-type networks. For the BNN, implementation and experimentation were carried out using the PyMC 5 library [53], which enables the construction of Bayesian models with a Python API and fitting these models using Markov chain Monte Carlo (MCMC) techniques.

Each signal was assigned a class: signals from a non-damaged device belong to class 0, while the remaining signals belong to class 1. To verify the accuracy of the developed Bayesian network, the dataset was randomly divided into training and test sets in a 50/50 proportion.

Figure 2 presents the process of an experiment aimed at comparing the performance of classical Bayesian networks and deep neural networks in the task of classifying the technical condition of an electric device. The dataset used was obtained from [50] and includes audio recordings categorized into two main classes: (1) healthy devices and (2) devices with a damaged motor. The second class is further subdivided into four fault types: damaged gear train (Fault 1), five broken rotor blades (Fault 2), ten broken rotor blades (Fault 3), and a shifted brush (motor off) (Fault 4). Each class consists of 30 one-second-long recordings stored in WAV format with a sampling rate of 4400 Hz.

The dataset was randomly divided into training and testing subsets in a 50/50 ratio. During preprocessing, the signals were prepared for model training. Two parallel experiments were then conducted: one using classical Bayesian networks and the other using deep neural networks. After classification, the results were collected, and performance metrics such as accuracy, sensitivity, specificity, and F1-score were computed. The goal of the experiment was to evaluate which model architecture performs better in classifying audio signals of electric devices with various fault conditions.

The analysis was repeated 1000 times. The average accuracy of the network was approximately 80% for BNN and 75% for DNN. The results of the BNN operation are illustrated in Figure 5. An example confusion matrix for DNN is presented in Figure 6 and for BNN in Figure 7. The displayed tables demonstrate the performance of the developed BNN. The suggested design detects damage flawlessly (top-right) and accurately identifies signals from undamaged devices in nearly 80% of instances. This is a desirable characteristic, as it informs about malfunctions in the system with high accuracy. Looking at the DNN confusion matrix in Figure 6, we can also see that the good result indicated by DNN is misleading due to the poorly balanced training set, in which the class of damaged devices dominates. This is typical for this type of dataset; however, we see here the advantage of the BNN solution, in which we do not need to balance the training set.

Table 3 presents a comparative analysis of two neural network models (Bayesian neural network (BNN) and recurrent neural network (RNN)) in the context of engine fault identification. The models were evaluated using four standard performance metrics: accuracy, precision, recall, and F1-score, each reported with their mean values and standard deviations to reflect both effectiveness and consistency.

The accuracy of the BNN model reached 80% with a low standard deviation of 1%, indicating not only high classification performance but also stability across multiple runs. In contrast, the DNN model achieved a lower average accuracy of 75.4%, accompanied by a slightly higher standard deviation of 1.4%, suggesting reduced and more variable performance.

When comparing precision, the BNN again outperformed the DNN, achieving 80% with a deviation of 1.5%, which implies the model makes fewer false-positive classifications. The DNN obtained a considerably lower precision of 55.2% ($\pm 1.1\%$), highlighting its tendency to generate more incorrect positive identifications.

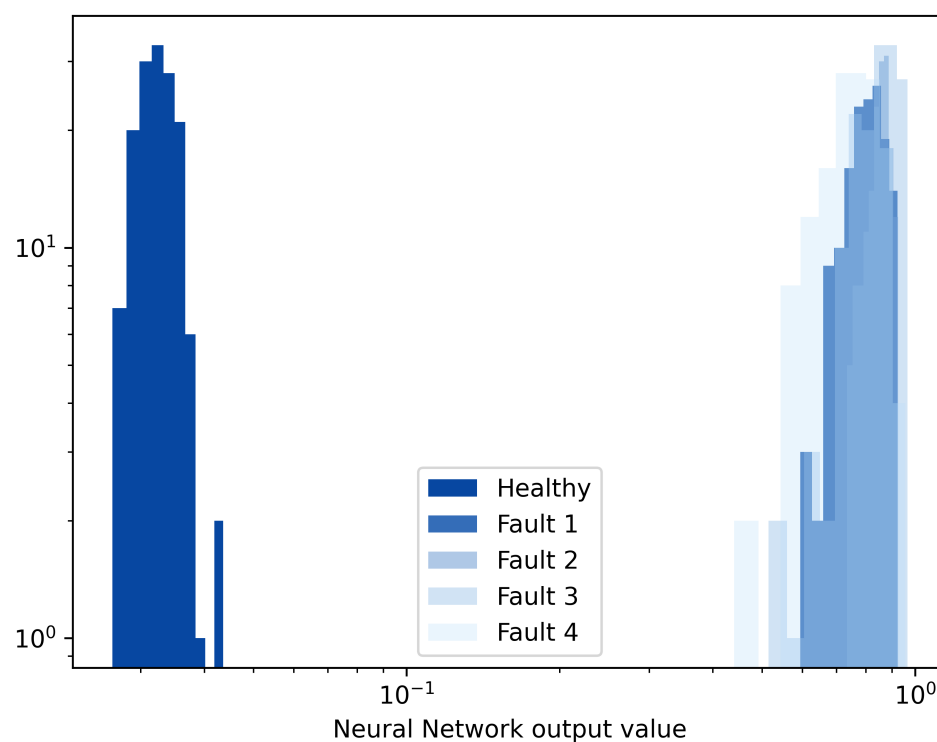


Figure 5. Comparison of the distributions of output values of the designed BNN for all signal classes.

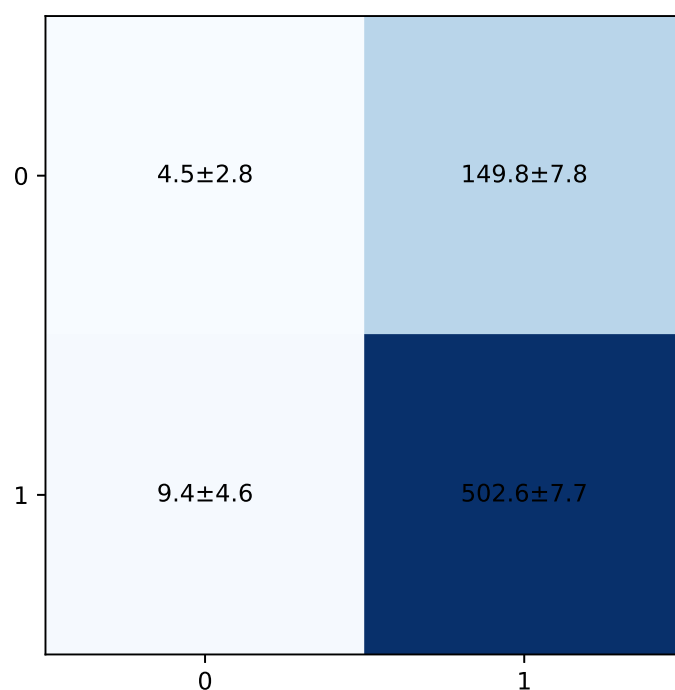


Figure 6. Statistical confusion matrix for one of the 1000 trials of the developed DNN. Values in all fields represent the average classification results along with the standard deviation. The proposed structure flawlessly detects damages and correctly recognizes signals from undamaged devices in almost 80% of cases.

The recall metric, which reflects the model's ability to detect actual faults, showed a similar trend. The BNN reached 71.9% with 1.1% deviation, demonstrating a strong capability to correctly identify a majority of existing engine faults. The DNN, however,

only achieved 50.5%, missing nearly half of the actual faults, despite having comparable variability ($\pm 1\%$).

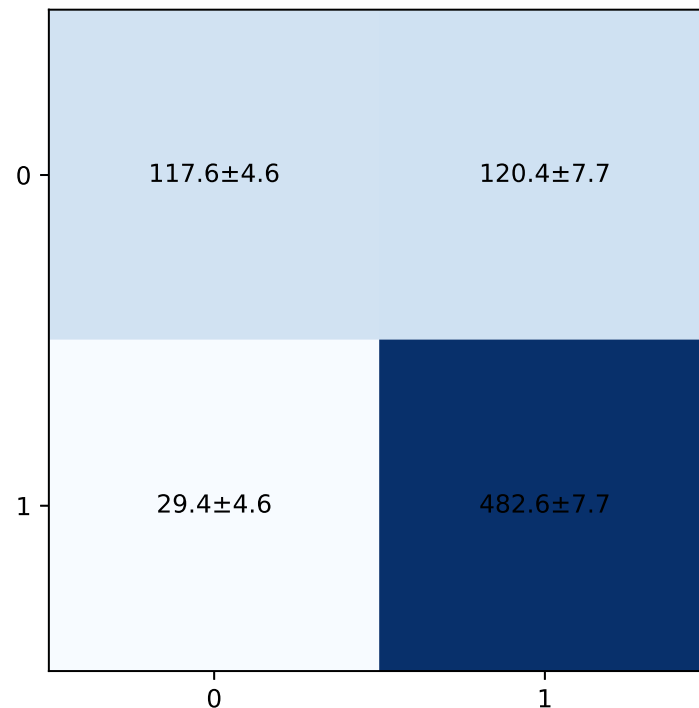


Figure 7. Statistical confusion matrix for one of the 1000 trials of the developed BNN. Values in all fields represent the average classification results along with the standard deviation. The proposed structure flawlessly detects damages and correctly recognizes signals from undamaged devices in almost 75% of cases.

The F1-score, which harmonizes precision and recall, further emphasized the advantage of the BNN model. With an F1-score of 73% ($\pm 1.2\%$), the BNN proved to be a balanced and reliable classifier. Meanwhile, the RNN scored only 45.6%, with a higher deviation of 1.7%, confirming its overall inferior performance in handling the fault classification task.

In summary, the Bayesian neural network consistently outperformed the recurrent neural network across all evaluated metrics. Not only did the BNN achieve higher average scores, but it also demonstrated greater robustness and lower variability, making it a more dependable choice for the engine fault identification process. These findings suggest that incorporating Bayesian inference in neural architectures can lead to improved predictive performance, especially in safety-critical applications like fault diagnosis.

Table 3. Statistical comparison of Bayesian neural network (BNN) and recurrent neural network (DNN) performance in engine fault identification. The table reports the mean and standard deviation (Std) of accuracy, precision, recall, and F1-score for each model over multiple evaluation runs.

	Accuracy		Precision		Recall		F1-Score	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std
BNN	80%	1%	80%	1.5%	71.9%	1.1%	73%	1.2%
DNN	75.4%	1.4%	55.2%	1.1%	50.5%	1%	45.6%	1.7%

Additionally, as shown in Figure 8, we can see that in BNN we can interpret statistical distributions of features, which is not possible in the DNN architecture.

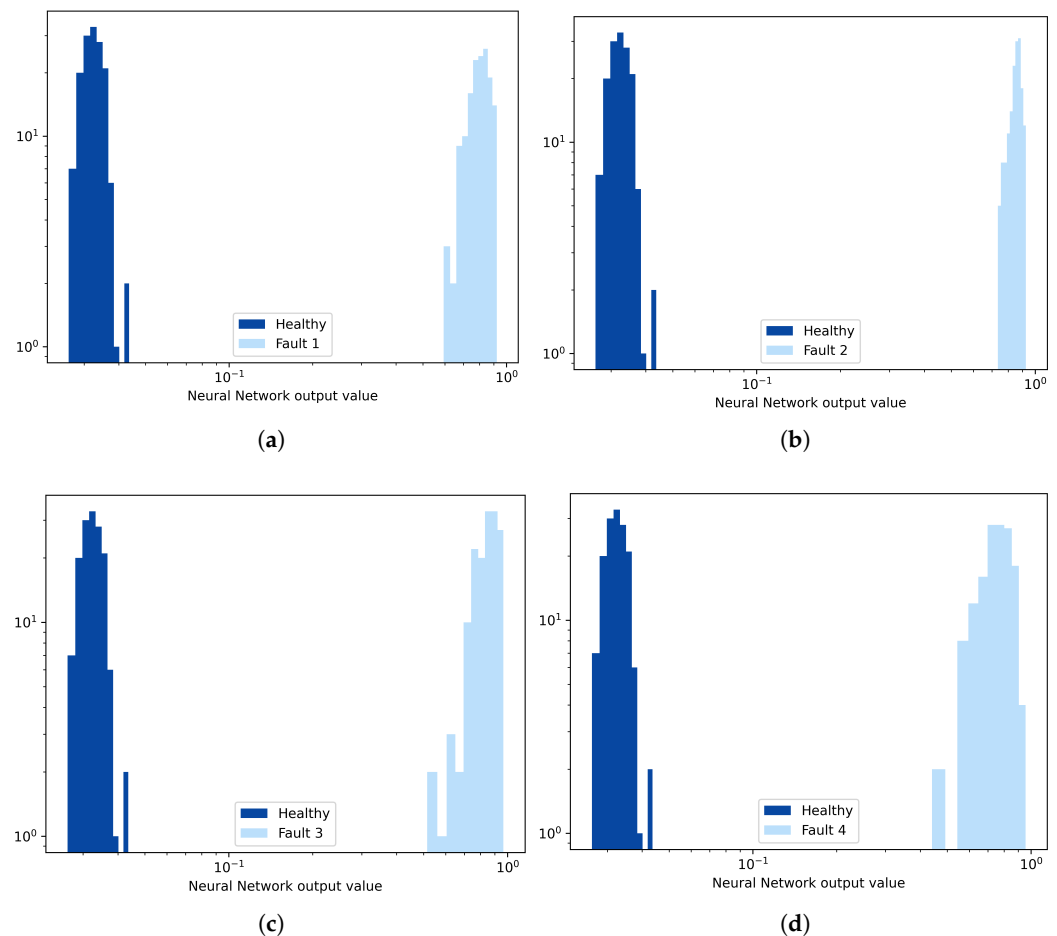


Figure 8. Comparison of the distributions of output values of the designed BNN for a healthy signal and individual classes of faults. (a) Comparison of the distributions of output values of the designed BNN for healthy and Fault 1 signals. (b) Comparison of the distributions of output values of the designed BNN for healthy and Fault 2 signals. (c) Comparison of the distributions of output values of the designed BNN for healthy and Fault 3 signals. (d) Comparison of the distributions of output values of the designed BNN for healthy and Fault 4 signals.

6. Conclusions

Bayesian neural networks (BNNs) and deep neural networks (DNNs) have been applied to fault detection. BNNs detect faults more accurately than traditional neural networks, even for imbalanced or incomplete datasets. In this paper, we focus on fault diagnosis of electric motors using acoustic signals. Such diagnostic signals in the time domain have varying lengths and are sampled at high frequency, thus necessitating a new signal representation for classification.

This study presented a comparative analysis of Bayesian neural networks (BNNs) and deep neural networks (DNNs) for detecting faults in electric motors using acoustic signals. The investigation focused on transforming time-domain sound signals into the frequency domain to facilitate consistent and informative input representations for classification tasks.

The experimental results clearly demonstrate that BNNs outperform DNNs across key performance metrics, including accuracy, precision, recall, and F1-score. Notably, the BNN achieved a mean accuracy of 80% with a lower variance, while the DNN lagged behind with 75.4% accuracy and significantly lower precision and recall. This indicates that BNNs not only offer higher classification accuracy but also provide greater robustness in handling imbalanced datasets, which is a common challenge in real-world diagnostic scenarios.

Beyond performance metrics, the probabilistic nature of BNNs introduces a valuable feature: the ability to quantify uncertainty in predictions. This capability enhances the interpretability and trustworthiness of diagnostic systems, particularly in safety-critical environments. By contrast, deterministic DNNs lack this feature, making them less informative in uncertain or noisy conditions.

Overall, the findings highlight the potential of Bayesian neural networks as a robust and interpretable solution for acoustic-based fault detection in electric motors. Future work will aim to explore the computational efficiency of various neural architectures and extend the approach to multi-class fault diagnosis. Developing standardized benchmarks and evaluating resource consumption during training and inference will also be crucial for practical deployment in industrial applications. Although the accuracy of the presented methods can be further improved by applying additional preprocessing and dataset balancing techniques, the primary objective of this study was to investigate the properties and comparative performance of different artificial neural network models. An open problem is the comparison of different architectures with respect to the resources needed for their training and classification. The next step would be to develop a set of benchmarks and metrics that would help compare different architectures. In addition, there are plans to attempt to develop classifiers capable of distinguishing between different classes of faults for different neural network architectures and to compare their performance.

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